Throughout this module, we were tasked with creating a water sensor, broken down into each component and process. To start, we cut masks in Wilbur Powerhouse with the vinyl cutter to create a stencil for an interdigitated electrode. The powerhouse had a plethora of useful machines, from 3D printers to a waterjet cutter, which we did not use but I found intriguing, and I plan on utilizing that lab throughout my time here. The next lab session, my group and I created several IDEs with the masks from Wilbur and silver paste on glass slides. One major difficulty that arose came with applying the masks to the slides. Because the masks had very thin arms, they ripped incredibly easily. We discovered that using an Xacto knife allowed us to be more accurate in removing the adhesive, and from there we were able to create three effective IDEs.

After the IDEs were constructed, we discussed materials used in computer chips throughout the market and the processes used to fabricate them. To explore this topic more in-depth, we toured the clean room facility where a microfilm photoresist was applied to a silicon wafer through spincoating to produce precision computer components. We then applied this concept to our water sensor, by applying a microfilm of PDOT, a polymer that alters its conductivity in a change in moisture, to the IDEs with spincoating. This essentially completed the physical component of the sensor. We then tested the sensor, by measuring the resistance of the device with and without the presence of water. Dry air caused the resistance to be low, while the presence of moisture caused the resistance to increase. The sensor worked well, and my group found the project to be a success.

Overall, I feel like my knowledge of sensors and computer manufacturing processes has grown with this module. The process of spincoating was particularly fascinating to me, especially when applied to photolithography. Most of the issues that arose over the project stemmed from making the masks for the electrodes. Because the tape required such precision by

hand, the masks were difficult to do correctly. However, my group and I found easier processes for applying the masks. On the other hand, I feel like my group had very strong teamwork skills. Across any issue that came up, we would talk it out and eventually come to a strong solution. I believe that this module allowed me to gain valuable knowledge in materials engineering, manufacturing, and computer engineering, which I will take with me into the rest of my engineering career.